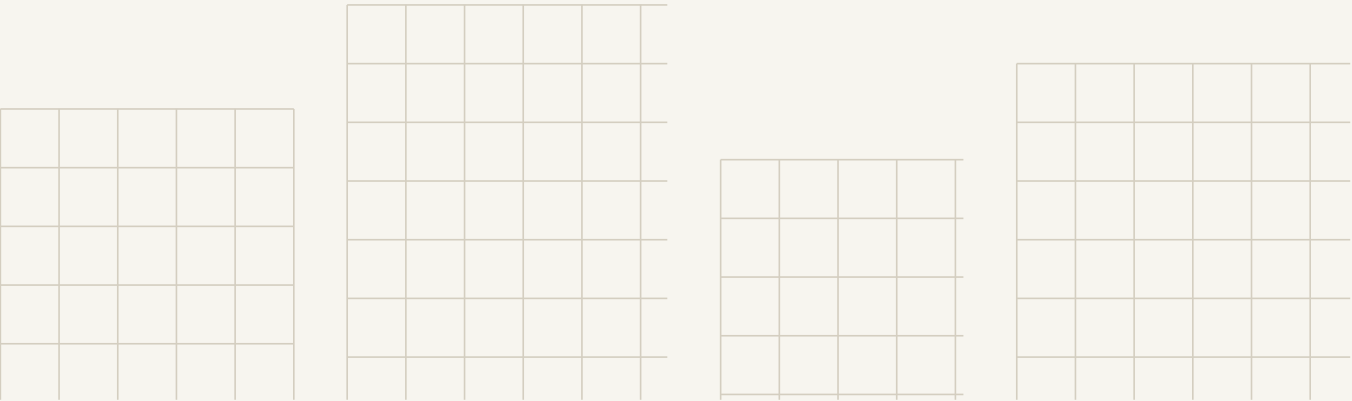


PROJECT PROPOSAL

Construction Site Monitoring System

AI-powered safety, productivity
and progress management

Single-site scope, implementation approach
and commercial framework



Prepared by SRS Logics

Software Development & Technology Solutions

Client / project: To Be Finalized

01 / EXECUTIVE OVERVIEW

Proposal overview

Camera-based monitoring for one construction site.

SRS Logics proposes a camera-based monitoring system for one construction site. As the client currently has no hardware, the budget includes cameras, recording and network equipment. The system will provide live views, agreed safety and activity alerts, and evidence for review. Additional sites are outside the present scope and can be considered separately.

Safety monitoring

Flag possible PPE violations and restricted-zone entries. Safety officers can review the footage and record the action taken.

Site activity

Record visible workforce presence, vehicle movements, equipment activity and agreed material-handling events.

Progress records

Keep dated images from fixed cameras so project teams can compare changes and review visible milestones.

Management reporting

Bring site reports, open incidents and camera status into one dashboard, with access restricted by role and project.

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02 / PLATFORM SCOPE

Core monitoring & operations

The following modules form the proposed scope. All functions are subject to site testing.

Site setup and zones

Configure the project and its site, buildings, towers, floors, gates, zones and camera locations. Show workforce observations, safety events, incidents and camera health for this site.

Live cameras & infrastructure health

Site/zone listings, live view, camera grids, search, full-screen access and event timelines. Track offline cameras, unavailable streams, interrupted connections and processing failures.

Workforce & presence

Person detection, visible counts, zone occupancy, movement and entry/exit observations. Working-hour estimates and named attendance depend on approved identification integrations.

Safety, PPE & restricted zones

Configure helmet/vest detection and visible PPE where practical; zone intrusion and unsafe-area rules. Congestion, person-down and smoke/fire indicators require separate feasibility validation.

Vehicles, gates & machinery

Detect and classify selected cars, trucks, dumpers, mixers, excavators, loaders and cranes. Record gate entry/exit and movement events; optionally validate ANPR at suitable gates.

Materials & equipment activity

Observe loading/unloading, material arrival/departure and storage-area movement. Track visible equipment presence and approximate activity periods; integrate inventory or telematics for precision.

03 / INTELLIGENCE & ACCESS

Progress, reporting & connected systems

Progress records, reports and access for site teams and management.

Visual construction progress

Scheduled fixed-camera captures, daily/weekly images, before-and-after comparisons, timeline views, milestone evidence and time-lapse. Advanced change detection, structural-stage, floor-level or area-completion indicators require clearly defined, measurable use cases.

Analytics & daily reports

Workforce observations, peak occupancy, zone activity, PPE and incident trends, vehicle movements, equipment activity, camera health and progress summaries. Filter by project, site, camera, zone, date, event type and severity; export formats to be agreed.

Web, PWA & role access

Responsive dashboard and mobile/PWA access for cameras where supported, alerts, incident review, progress and reports. Roles: super administrator, management, project manager, site manager, safety officer, security and read-only viewer.

OPTIONAL INTEGRATIONS

Operational need	Potential integration
Identity & access	RFID, QR/ID cards, approved biometric attendance and access control
Materials & inventory	QR/barcodes, RFID, weighbridges, gate-pass, inventory and ERP systems
Equipment & site context	GPS, equipment telematics, IoT and environmental sensors
Project coordination	Project-management systems, BIM, existing CCTV/NVR and supported external APIs

Productivity indicators describe visible site activity. They require context and should support management review rather than determine individual performance or certified project completion.

04 / REFERENCE ARCHITECTURE

System architecture

We will select edge, cloud or hybrid processing after assessing the cameras and network.

01 Site cameras & existing CCTV/NVR

Compatible video streams from mapped gates, work zones, storage areas and progress viewpoints.

02 Site / edge processing

Stream ingestion, selected AI inference, event rules and optional local buffering. Local processing during outages depends on the selected design.

Alternative: cloud processing

Cloud/GPU inference where connectivity and capacity permit. Edge, cloud or hybrid topology is selected per site and use case.

EVENTS + METADATA + REQUIRED MEDIA

03 Central platform & event services

Python APIs and business logic; rules validation, incident ownership, notifications, reporting and integrations. Separate structured records from media evidence storage.

04 Data & evidence

Relational database; snapshots, available event clips and progress images in object/media storage.

05 Web / PWA access

Role-scoped live views, alerts, incident review, site analytics and reports.

Sizing the system

Sites and cameras; resolution and analyzed frame rate; model workloads; latency needs; bandwidth; retention; availability and data-location requirements.

Technology direction: responsive HTML/CSS/JavaScript frontend, Python services, OpenCV and task-specific detection models. Frameworks, model licenses and infrastructure providers: To Be Finalized.

05 / OPERATING WORKFLOW

Alerts and incident handling

Each alert should reach a named person and retain a record of the response.

01

Observe

Read the configured camera stream and identify a supported object or event.

02

Validate

Apply zones, schedules, confidence thresholds and event rules; suppress duplicate alerts where configured.

03

Preserve

Create an event with site, zone, camera, timestamp, event type and priority. Store a snapshot and available clip/reference.

04

Notify & assign

Route to the responsible safety, site or security team. Track severity, ownership and acknowledgement.

05

Review & resolve

Review evidence, confirm or dismiss the event, document action and comments, and close the incident.

06

Review & report

Keep the action history and include confirmed events in reports. Review recurring false alerts before changing rules or models.

Example: a possible missing helmet

Configured PPE event → safety officer review → site manager escalation if unresolved → management escalation for defined critical conditions.

Channels may include dashboard, PWA, email, SMS, WhatsApp or other supported APIs. Providers, delivery dependencies, escalation timing and recurring charges: To Be Finalized.

06 / RECOMMENDED ROADMAP

Implementation sequence

We recommend the following sequence. Dates and durations are To Be Finalized.

- | | | |
|----|---|-----------------------------------|
| 01 | Technical discovery / site survey
Agree use cases, locations, stakeholders and constraints. | Approved discovery scope |
| 02 | Camera / network assessment
Plan camera coverage, stream requirements, lighting, power and connectivity. | Readiness and remediation plan |
| 03 | Proof of concept
Evaluate selected use cases with representative site footage and cameras. | Documented POC results |
| 04 | AI accuracy validation
Measure errors and operational performance against agreed acceptance criteria. | Acceptance or further testing |
| 05 | Core platform
Build site/camera setup, role access, dashboard, validated AI events and incidents. | Platform review and acceptance |
| 06 | Pilot deployment
Operate at the selected site, train users and validate end-to-end workflows. | Pilot review and support handover |
| 07 | Site production deployment
Move the accepted pilot into routine use at this site; confirm monitoring and support. | Site-level acceptance |
| 08 | Optional AI / IoT integrations
Extend vehicles/ANPR, materials, progress AI and connected systems as validated. | Acceptance of each added function |

The pilot should include the reports and mobile review needed by site users. Vehicle, equipment, material and progress modules can then be added in the agreed order. Additional sites are outside this proposal. Any optional AI function will be checked before operational use.

07 / PROOF OF CONCEPT

POC and acceptance testing

Start with one site, a selected camera set and a small number of agreed use cases.

Prioritize person detection, helmet/vest checks, restricted-zone events and selected vehicle detection. Confirm the final camera count, use cases and operating conditions during discovery. The POC establishes feasibility; the pilot validates the operational service with real users.

Validation area	Proposed method
Representative data	Sample day/night conditions, shifts, distances, occlusion, weather and normal/abnormal activity. Obtain authorized, labeled examples.
Independent evaluation	Separate model tuning material from held-out evaluation footage. Report results by use case, camera and operating condition.
Detection quality	Measure precision (correct alerts), recall (events found), missed events and false alerts per camera-hour. Document count error where relevant.
Operational performance	Measure end-to-end alert latency, stream availability, processing capacity, evidence retrieval and notification delivery.
User acceptance	Test alert assignment, review, escalation, closure, reporting and camera outages with nominated site users.

Acceptance criteria

Targets, sample size, test duration, acceptance owners and allowed operating conditions: To Be Finalized before testing.

Decision after testing

Accept proven use cases, refine camera placement or models, or defer unsupported scenarios. Record limitations before pilot approval.

08 / ASSUMPTIONS & LIMITATIONS

Assumptions and technical limits

The following conditions affect what can be delivered and how results should be used.

Area	Boundary / project assumption
Camera visibility	Resolution, angle, mounting height, distance, glare, low light, rain, dust, obstruction and motion affect detection. Blind spots remain outside coverage.
People & attendance	Visible counts are observations, not a unique site headcount. Occlusion and cross-camera duplication can affect counts. Formal named attendance needs reliable identification.
PPE & critical safety	Only visually discernible PPE can be assessed. Small items may be unsuitable. Fall/person-down and smoke/fire indicators need separate validation and human review.
Vehicles & equipment	ANPR depends on plate visibility, lighting, motion and gate geometry. Visual movement is not engine run-time, fuel use, diagnostics or precise GPS location.
Materials & progress	Video alone does not establish exact stock quantity, weight, hidden work quality or certified completion. Inventory, weighbridge, schedule or BIM data may be required.
Infrastructure & availability	New cameras and network equipment are required. Client arranges site access, power and internet readiness. Outages can interrupt cloud alerts; buffering/recovery must be designed and tested.
Model maintenance	No universal accuracy guarantee. New camera positions, site stages, PPE or equipment can require recalibration, new labels, custom training and revalidation.

Site safety responsibilities

AI supports review and response. It does not replace supervision, mandated safety controls or dedicated fire and life-safety systems.

09 / PRIVACY & SECURITY

Privacy and security

These controls will be agreed with the client before production deployment.

Purpose & transparency

Define monitoring purposes, permitted use and responsible owners. Arrange appropriate worker/visitor notices and approvals. Identity or biometric integrations require separate assessment and authorization.

Access & accountability

Apply least-privilege roles and project/site restrictions; protect administrative accounts, sessions and credentials. Log access, incident actions, evidence exports and configuration changes.

Secure architecture

Encrypt data in transit and at rest where supported. Segment camera networks, protect secrets and restrict external access. Review hosting, APIs, model licenses and service-provider access.

Retention & minimization

Agree retention separately for raw video, snapshots, event clips, progress records and logs. Limit transfer to necessary data; define deletion, export, backup and incident-preservation rules.

Human review & responsible use

Retain evidence for authorized review, allow false-event dismissal and capture corrections. Avoid unsupported conclusions about individuals from activity signals; document model limits for users.

Operational resilience

Define backup/restore, monitoring, patching, vulnerability handling and security-incident response. Set availability, recovery and support targets, then validate them for the chosen deployment.

Client-specific data location, retention periods, legal requirements, identity permissions and security acceptance criteria: To Be Finalized with the client and its designated reviewers.

10 / DELIVERY & OWNERSHIP

Deliverables and responsibilities

The final statement of work will confirm the deliverables for each approved stage.

Stage	Expected deliverables
Discovery & assessment	Requirements and prioritized use cases; site/camera inventory; coverage and network assessment; readiness actions; proposed deployment architecture.
POC & validation	Configured POC; labeled evaluation approach; measured AI and infrastructure results; limitations register; recommendation on which functions to proceed with.
Core & pilot	Responsive web/PWA platform; role and site setup; compatible cameras; validated AI modules; alerts, incidents, evidence, camera health and agreed reports.
Deployment & handover	Accepted single-site configuration; agreed integrations; deployment/configuration documentation; user/admin guidance; training and operational handover.
Advanced scope	Separately agreed models and integrations for ANPR, materials, progress AI, attendance, ERP/BIM, GPS, sensors or equipment telematics.

WORKING RESPONSIBILITIES**SRS Logics**

Solution design, agreed software and AI configuration, integration development, testing, deployment assistance, documentation and training within the finalized scope.

Client team

Nominate owners and reviewers; provide authorized site, camera, network and representative data access; coordinate new hardware installation, workforce communication and acceptance feedback.

Source code/IP ownership, model and third-party licenses, hardware procurement, training coverage, warranty, maintenance and support arrangements: To Be Finalized.

11 / COMMERCIAL PROPOSAL

Development in three payments

One site. Eight cameras assumed. Payments linked to the agreed delivery milestones.

DEVELOPMENT AND TESTING

Total fee: Rs. 5,00,000

The development fee is paid in stages, with the final payment after handover.

At contract signing**Rs. 1,50,000**

30% of development fee

After development**Rs. 2,00,000**

40% of development fee

After handover**Rs. 1,50,000**

30% of development fee

Hardware, installation and monthly services are separate expenses, detailed below and on page 13. The signing payment shown above is the development instalment only. Applicable taxes are additional.

SITE SETUP - A SEPARATE, ONE-TIME EXPENSE

Item	Budget allowance
IP cameras - eight units	Rs. 40,000-64,000
Recording equipment and surveillance storage	Rs. 25,000-40,000
Network switches, routers and UPS	Rs. 20,000-35,000
Cabling, mounting, installation and configuration	Rs. 20,000-40,000
Site setup allowance - total	Rs. 1,05,000-1,79,000

Confirm the site requirements first, then agree the equipment quotation and procurement terms before ordering. The amounts above are budget allowances; final costs depend on the survey and supplier quotations.

12 / OPERATING COSTS & TERMS

Services to run the site

Monthly services cover the AI processing, application, event storage and connectivity.

MONTHLY SERVICE ALLOWANCE

Service	Monthly allowance
AI processing, hosting, database and event storage	Rs. 35,000-50,000
Site internet connection	Rs. 1,000-3,000
Total per month - one site	Rs. 36,000-53,000

Budget Rs. 36,000-53,000 per month, separate from development and site setup. This is an estimate; actual charges depend on services, capacity and usage. Billing commencement will be agreed before provisioning. Applicable taxes are additional.

BASIS OF ESTIMATE

Cameras and coverage

The client currently has no hardware. Eight cameras is a budget assumption; the survey will confirm coverage, equipment specifications, cable routes and installation requirements.

Cloud processing

The allowance includes a dedicated cloud AI server and application services. Selected AI checks use reduced frame rates. Continuous footage stays on site. Capacity and retention will be validated in the POC.

ADDITIONAL ITEMS, IF REQUIRED

Dedicated ANPR cameras, extensive outdoor cabling, poles or civil work, advanced sensors and integrations, and ongoing software maintenance are outside these allowances. Any separate discovery, custom training, data-labeling or third-party charges must be agreed before work begins.

CONTRACT AND DELIVERY TERMS

Delivery dates and durations for discovery, POC, development and deployment: To Be Finalized. The contract will define development completion and handover criteria. Support, warranty, procurement responsibility and proposal validity: To Be Finalized.